Predicting Delivery Time for Olist

Can we give customers a more accurate delivery estimate?

A REGRESSION PROJECT USING OLIST'S BRAZILIAN E-COMMERCE ORDER DATA

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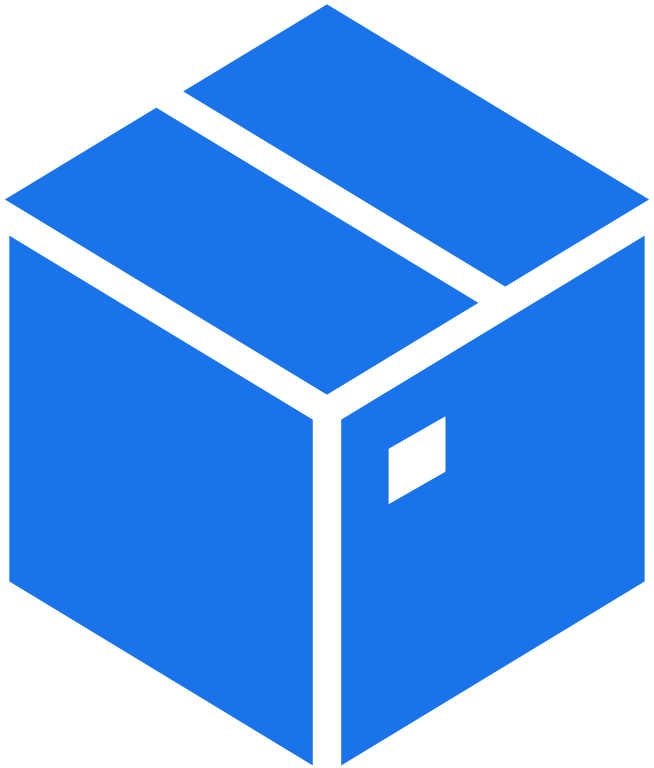


Problem Statement

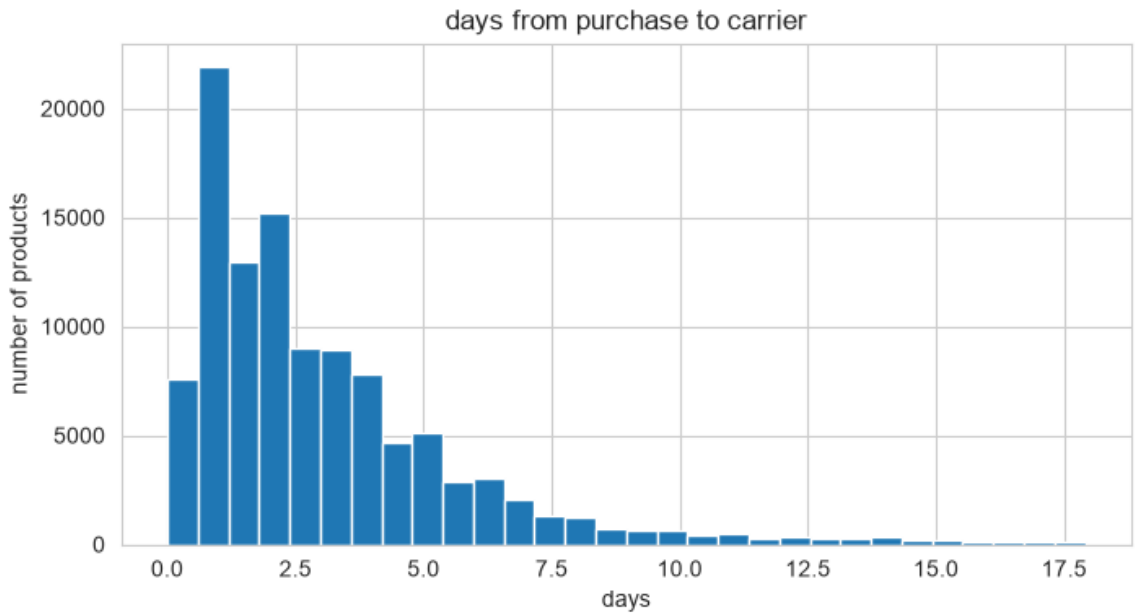
- Olist often gives customers delivery estimates that are much longer than the actual delivery time. On average, the estimated time is 23.7 days, while orders arrive in 12.6 days. These long estimates may discourage customers from completing their purchases. Our goal is to predict a more accurate delivery time using information available when an order is placed.

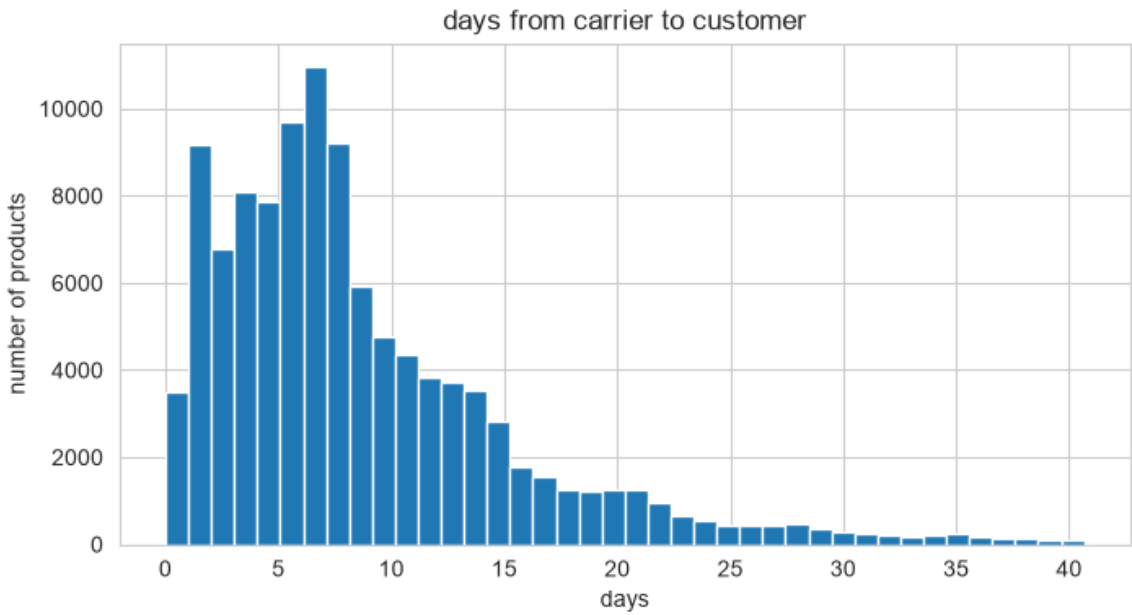
Our Objective

- We aim to predict the number of days between purchase and delivery using order details, product information, location, and timing. A more accurate estimate could help customers plan better and help Olist improve its delivery promises.



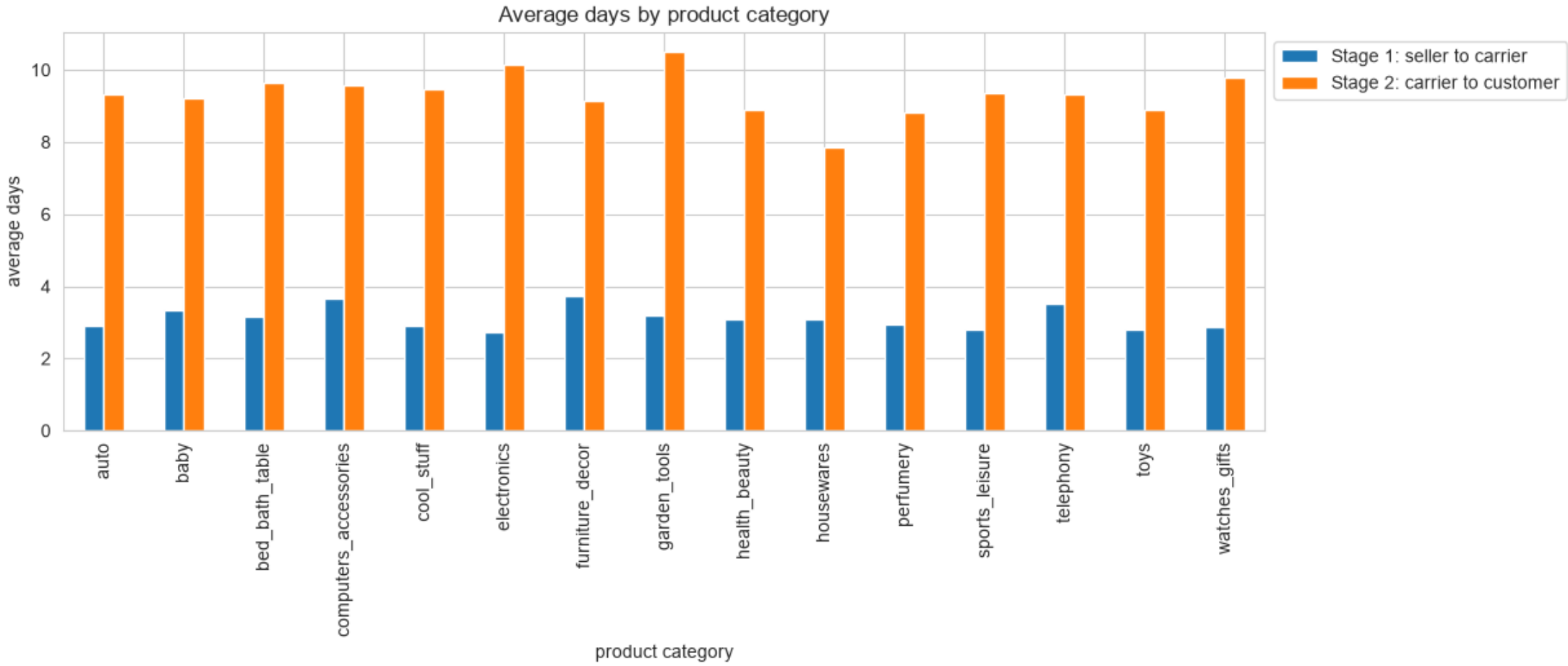
Exploring Delivery Times





Median: 2.2 days to carrier | 7.1 days from carrier to customer

Delivery Time by Product Category



The carrier-to-customer stage takes longer than seller-to-carrier across these categories.

Delivery Time by Region

- Location makes a clear difference. Orders in **São Paulo** take about 8.8 days on average, compared with 23.8 days in **Pará** and 26.4 days in **Amazonas**. Because delivery conditions vary across regions, one fixed national estimate may not work well for everyone.

State	Orders	Real days	Promised days
São Paulo (SP)	40,494	8.8	19.1
Minas Gerais (MG)	11,354	12.0	24.5
Pará (PA)	946	23.8	37.2
Amazonas (AM)	145	26.4	45.3

Target and Features

- Our target is **delivery_days**: the time from purchase to delivery, measured in days.
- We used features such as customer location, seller location, purchase month, freight cost, and product details. We avoided using information that would only become available after the order is delivered.

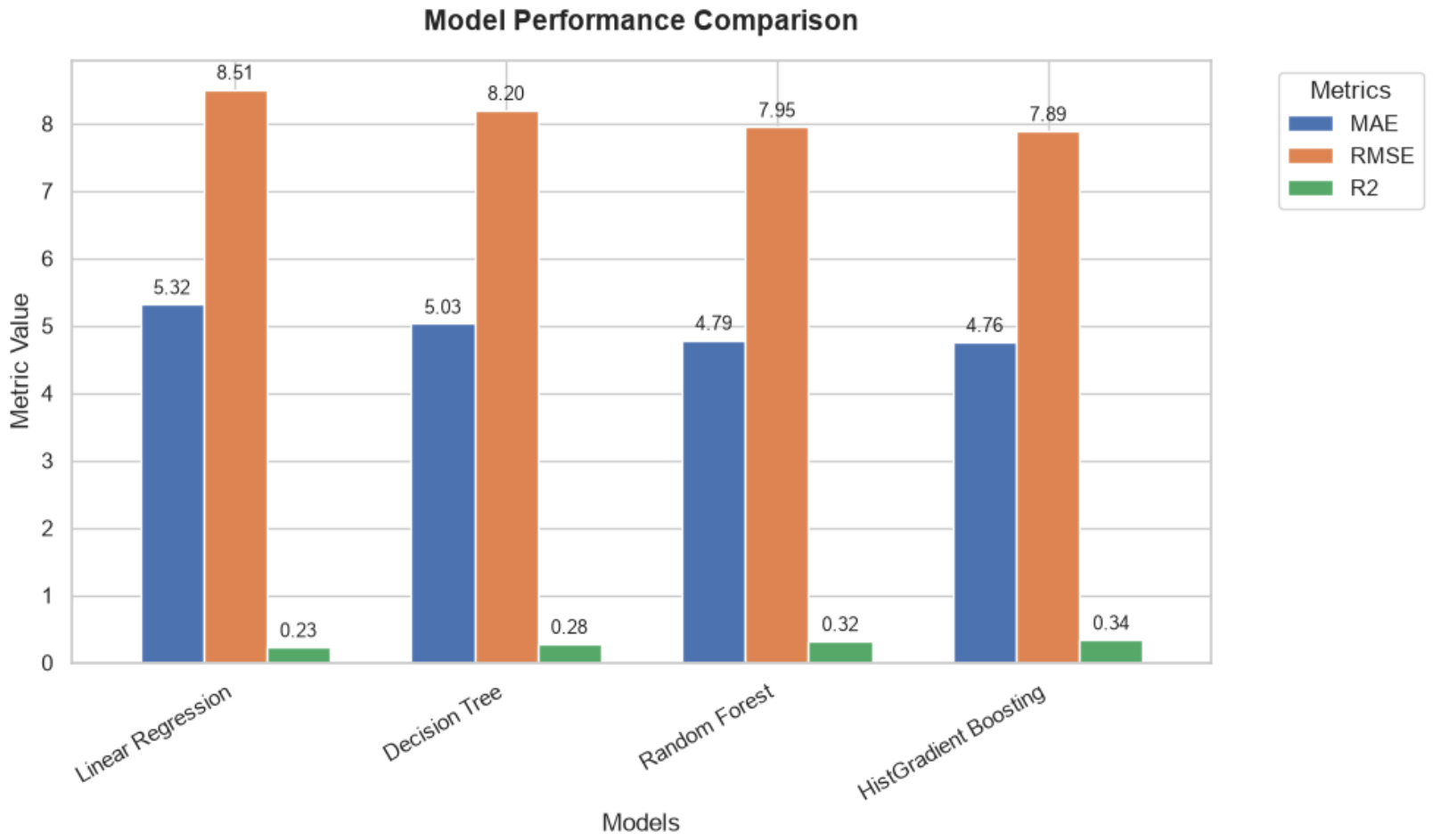
Preparing the Data

- We split the data into training and test sets. Missing numerical values are filled with the median, and missing categorical values with the most frequent category. We used one-hot encoding for categories and scaling for Linear Regression.

Models and Evaluation

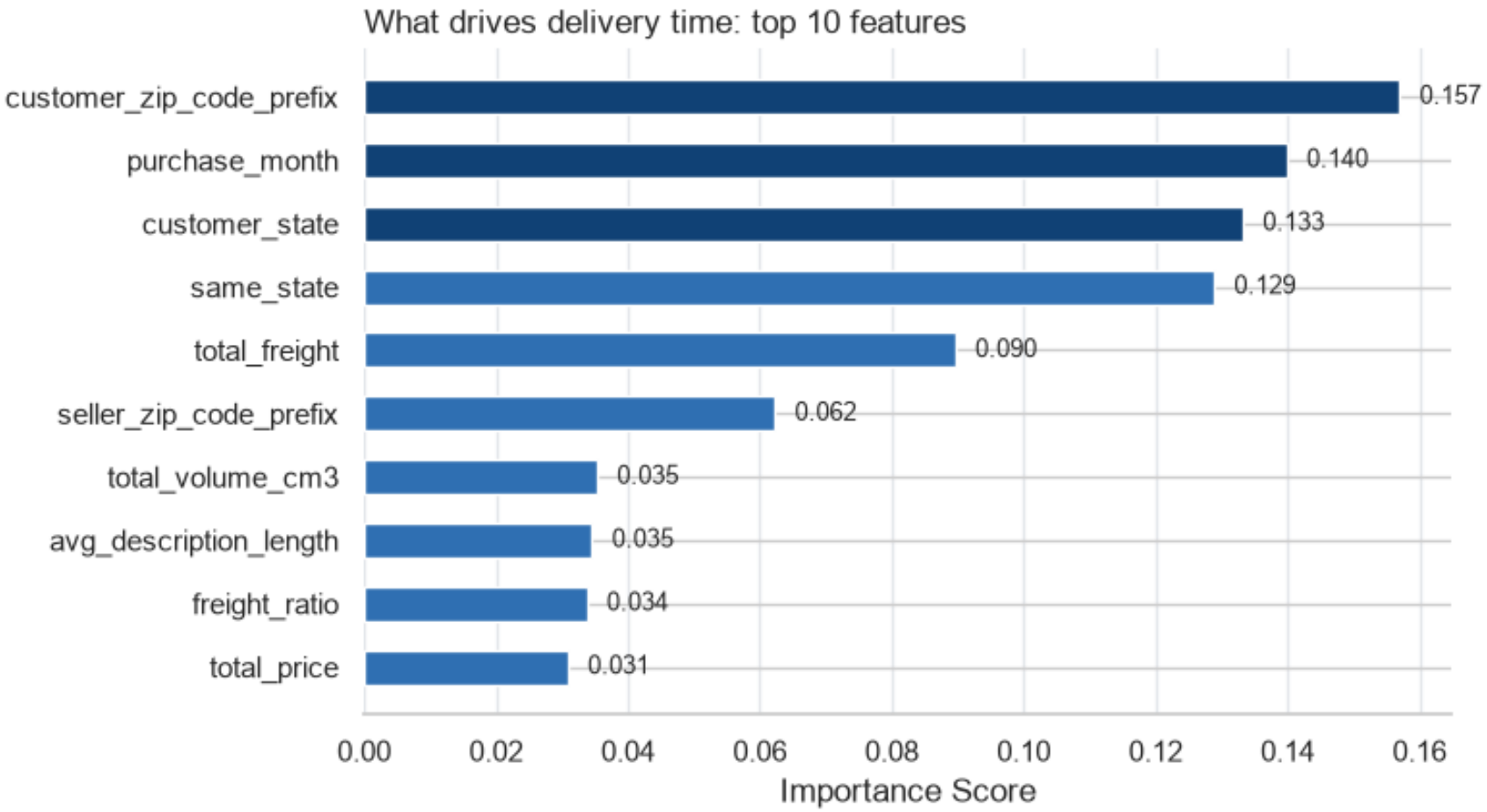
- We trained four regression models: Linear Regression, Decision Tree, Random Forest, and Histogram Gradient Boosting.
- Olist's current delivery estimate is used as a benchmark.
- We compared them using three evaluation metrics:
  - MAE (Mean Absolute Error): The average prediction error in days. Lower is better.
  - RMSE (Root Mean Squared Error): Gives more importance to large prediction errors. Lower is better.
  - R^2 (R-squared): Measures how much variation in delivery time the model explains. Higher is better, with 1 being a perfect score.
- Example: If Model A has an MAE of 4 days and Model B has an MAE of 6 days, Model A makes more accurate predictions on average.

Comparing Model Performance



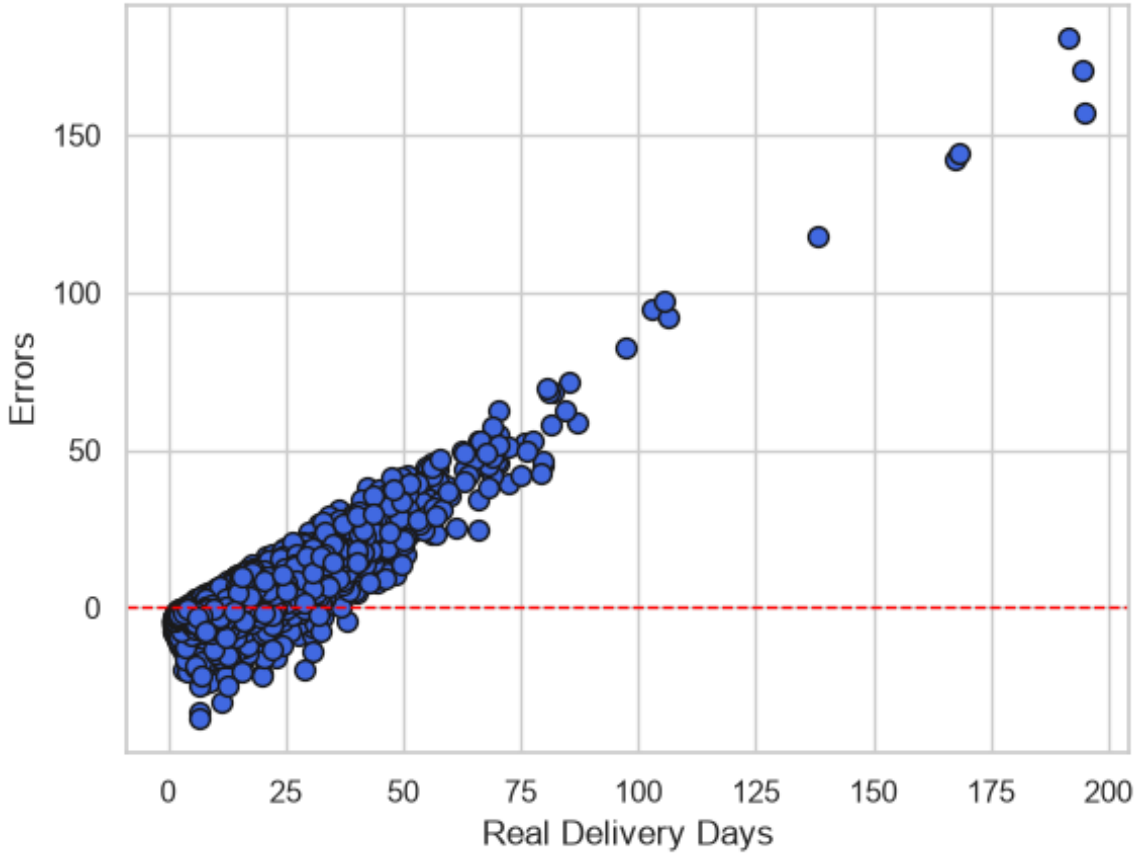
Histogram Gradient Boosting has the lowest validation MAE: 4.76 days.

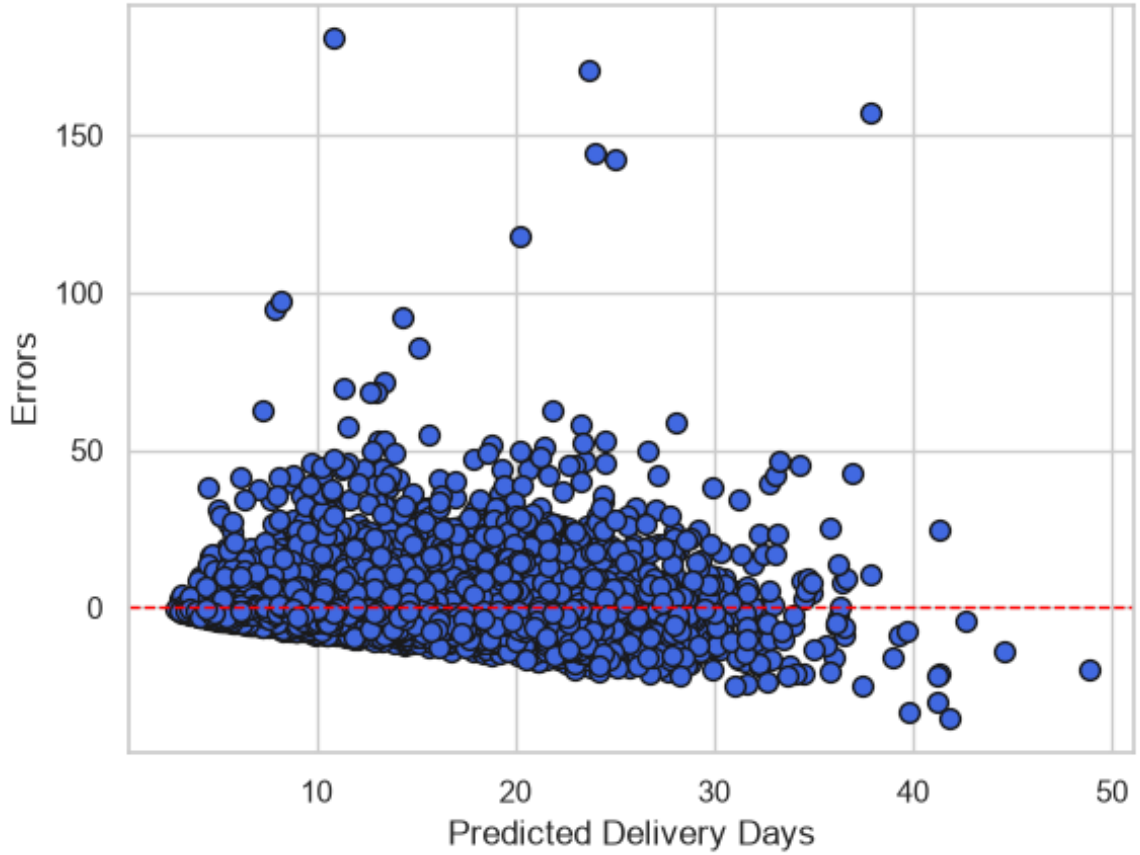
What Influences Delivery Time?



Location and purchase timing are among the strongest predictive signals.

Where Does the Model Make Mistakes?





Understanding Errors

- The model is unbiased on average (mean residual ≈ 0.1 days), but the median residual of −1.3 days shows it over-predicts typical orders slightly and under-predicts a small number of very slow ones. the model plays it safe on normal orders and misses the long tail.
- Errors grow with the predicted value: the typical error is about 2.4 days for predictions under 6 days and about 9 days above 20.

Limitations

The model is limited by the information available in the dataset.

Some extreme delays are likely caused by:

- Carrier disruptions
- Local disruptions
- Other operational factors not represented in the available features

The model therefore cannot fully anticipate certain unusual delivery events.

Final Results and Conclusion

- On the test set, our final model achieved an MAE of 4.68 days. This suggests that machine learning can provide more accurate delivery estimates than Olist's current approach, although extreme delays remain difficult to predict.

Thank You!

